

MWTH 301 Wi-Fi Temperature & Humidity Sensor Datasheet

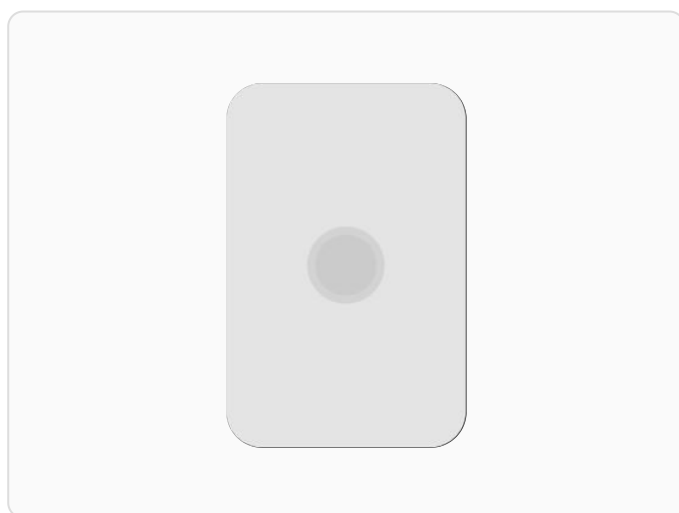
Product Overview

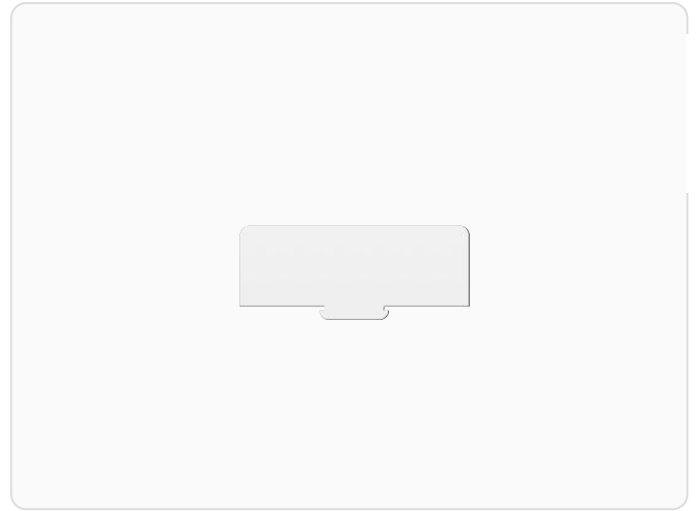
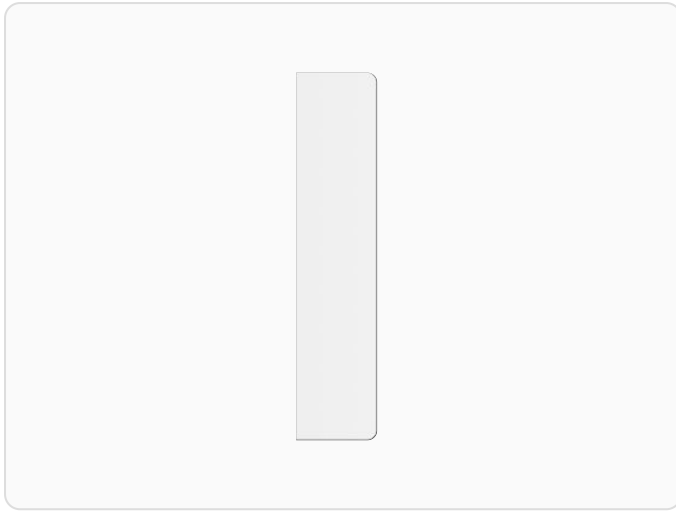
MWTH 301 is a compact, **low-power Wi-Fi temperature and humidity sensor** designed for accurate indoor and semi-industrial environmental monitoring where direct IP connectivity is required. It enables real-time climate data transmission over standard Wi-Fi networks without the need for gateways or long-range network infrastructure.

Built using industrial-grade sensing elements, MWTH 301 delivers fast response, low drift, and stable temperature and humidity measurements over long operating periods. Its optimized low-power architecture supports battery-operated deployments using intelligent sleep-wake cycles, making it suitable for continuous monitoring with minimal maintenance.

The sensor securely transmits data to **local servers, on-premise dashboards, or cloud platforms** using standard protocols such as **MQTT, HTTP, and HTTPS**. Local BLE support allows easy provisioning, diagnostics, and configuration through the Macnman Maya mobile application.

Housed in a rugged, ingress-protected enclosure, MWTH 301 is ideal for **smart buildings, HVAC systems, cold rooms, warehouses, laboratories, and industrial indoor environments** where reliable Wi-Fi connectivity, low latency, and accurate environmental data are essential.





Key Features

High-Accuracy Temperature & Humidity Sensing

Precision temperature and humidity measurements using industrial-grade sensing elements for reliable environmental monitoring.

Direct IP-Based Communication

Transmits data directly to local servers or cloud platforms over Wi-Fi without gateways.

Battery-Operated Design

Enables flexible deployment in locations where wired power is unavailable.

Local Configuration via BLE

Easy provisioning, diagnostics, and configuration using the Macnman Maya mobile application.

Compact & Lightweight Form Factor

Small enclosure suitable for wall or ceiling mounting in indoor environments.

Smart Data Reporting

Supports time-based, threshold-based, and event-driven reporting to optimize power consumption.

Low-Power Wi-Fi Connectivity

Optimized Wi-Fi communication with intelligent sleep-wake cycles for battery-efficient operation.

Standard IoT Protocol Support

Supports MQTT, HTTP, and HTTPS for easy integration with dashboards, BMS, and IoT platforms.

Fast Response & Long-Term Stability

Low sensor drift and stable performance over extended operating periods.

Secure Wireless Communication

WPA2 / WPA3 Wi-Fi security with TLS/SSL encrypted data transmission.

Ingress-Protected Enclosure

IP-rated housing designed for reliable operation in industrial and commercial indoor environments.

Smart Building & HVAC Ready

Ideal for climate monitoring, comfort analysis, and energy optimization applications.

Suited Applications

Smart Buildings & Offices

Continuous indoor temperature and humidity monitoring for comfort optimization, energy efficiency, and building management systems.

Cold Rooms & Cold Chain Storage

Reliable environmental monitoring for cold rooms, vaccine storage, food warehouses, and pharmaceutical storage areas.

Laboratories & Clean Rooms

Accurate climate monitoring for research labs, testing facilities, and controlled indoor environments.

Data Centers & Server Rooms

Tracks temperature and humidity to protect IT infrastructure and ensure operational reliability.

Hospitals & Healthcare Facilities

Supports temperature and humidity monitoring in wards, pharmacies, equipment rooms, and storage areas.

HVAC Monitoring & Optimization

Tracks temperature and humidity across HVAC zones to improve performance, efficiency, and preventive maintenance.

Warehouses & Storage Facilities

Ensures controlled temperature and humidity conditions for FMCG, electronics, raw materials, and finished goods.

Industrial Indoor Areas

Monitors environmental conditions inside factories, production floors, and utility rooms with existing Wi-Fi coverage.

Retail & Commercial Spaces

Maintains indoor comfort and monitors climate conditions in malls, stores, and commercial buildings.

Technical Specifications

General Specifications

Part Order Number	MWTH 301
Product Brand	Macnman Technologies Pvt Ltd

Temperature Measurement

Range	-30°C to +70°C
Accuracy	0°C to +70°C: ±0.3°C & -30°C to 0°C: ±0.6°C
Resolution	0.1°C
Calibration	Factory calibrated
Probe Options	Internal & External Both Options

Humidity Measurement

Range	0% to 100% RH
Accuracy	10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH
Resolution	0.5% RH
Calibration	Factory calibrated

Pressure Measurement*

Range	300 hPa to 1100 hPa
Accuracy	± 1.0 hPa (typical at 25°C, 300–1100 hPa)
Resolution	0.1 hPa
Long-Term Stability	$\leq \pm 1$ hPa/year
Calibration	Factory calibrated

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network

Single Red Blink	A confirmed parameter of MacSync is being modified
------------------	--

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based
Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 µA in sleep mode ~150 mA during transmission
Input Impedance	>10 MΩ for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MWTH 301 Variants

VARIANT NAME	POWER SOURCE	BATTERY CAPACITY	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MLTH 301-BO	Battery	09 Ah			Remote or portable deployments with low power

What's in the Box?

Make sure box contains following items before installations



MLTH 30x



**Mounting
Clip***



**User manual &
Warranty Card**



**IP
Extender***



**Mounting
Screws**

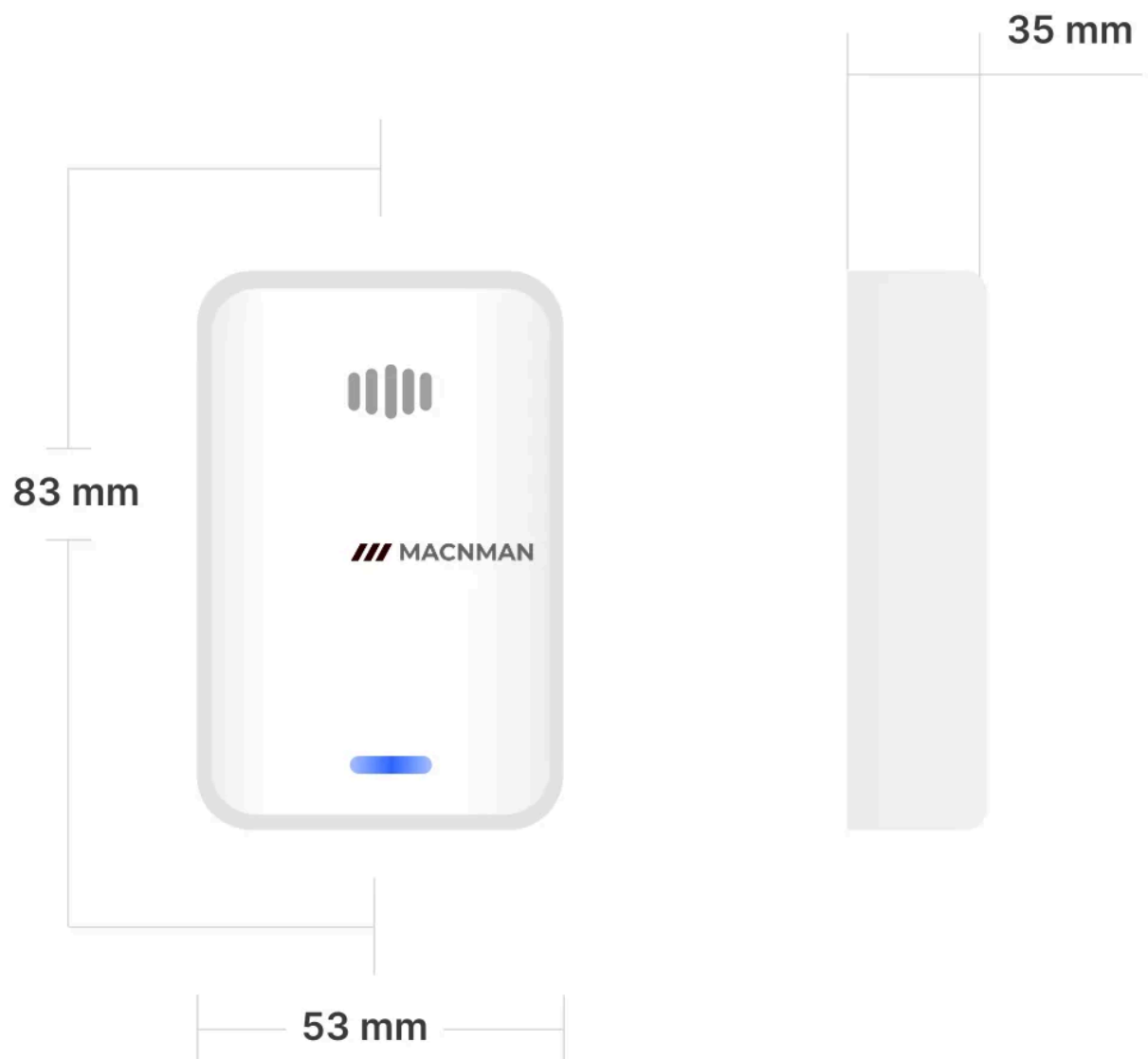


**Power
Adapters***

** Note**

Accessories must be purchased separately & may not be included in the box.

MWTH 301 Dimensions



Frequently Asked Questions

What is the MWTH 301 used for?

MWTH 301 is a compact, low-power, battery-operated Wi-Fi temperature and humidity sensor for indoor and semi-industrial environmental monitoring. Typical applications include Smart Buildings & Offices, HVAC Monitoring & Optimization, Cold Rooms & Cold Chain Storage, Warehouses & Storage Facilities, Laboratories & Clean Rooms and Industrial Indoor Areas.

What does the MWTH 301 measure, and how accurately?

Measurement specifications of the MWTH 301:

- Temperature: range -30°C to +70°C, accuracy 0°C to +70°C: $\pm 0.3^{\circ}\text{C}$ & -30°C to 0°C: $\pm 0.6^{\circ}\text{C}$, resolution 0.1°C.
- Humidity: range 0% to 100% RH, accuracy 10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH, resolution 0.5% RH.
- Pressure: range 300 hPa to 1100 hPa, accuracy ± 1.0 hPa (typical at 25°C, 300–1100 hPa), resolution 0.1 hPa.

How does the MWTH 301 connect and send data over Wi-Fi?

The MWTH 301 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band). Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

What is the IP rating and operating temperature of the MWTH 301?

The MWTH 301 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MWTH 301 configured?

The MWTH 301 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MWTH 301 work with?

The MWTH 301 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [Narmada TSX1 compact LoRaWAN temperature and humidity sensor](#), [MacSync-WT1 Wi-Fi temperature and humidity sensor](#), [MacSync-WRTDX2 Wi-Fi RTD PT100 temperature sensor](#), [MacSync-WRTDX5 Wi-Fi RTD PT1000 temperature sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)